On the Resistance of Screens

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Summary: The resistance of various screens in a tube is reduced to the drag of a single piece of wire by assuming that the average velocity in the screen itself rather than the free stream velocity is the significant factor. Almost all the available tests on screens or gauzes are brought into line with a single experimental function similar to the drag of a cylinder plotted against Reynolds number.

Notation

d	wire diameter
l	width of the square mesh
β	porosity = (area of holes)/(total area)
u_1, u_2	velocity before and after the screen
p_1, p_2	static pressure before and after the screen
ρ	density of fluid
ν	kinematic viscosity of fluid
k	resistant coefficient $= \dfrac{p_1 - p_2}{\frac{\rho}{2} u_1^2}$
c	reduced resistance coefficient $= \dfrac{\beta^2}{1-\beta} k$
c_D	drag coefficient of an infinite cylinder

1. Normal Flow Through Screens

The resistance of screens or gauzes in a tube has been measured several times since 1932[1-7]. Also some theories were established, two of which[2, 4] lead to formulae deriving the resistance only from the geometrical dimensions. Considering the complexity of the general problem of aerodynamic resistance, they are likely to be only rule-of-thumb formulae, even if they cover the tests in a certain range, and so a less ambitious approach might be preferable.

Attention is focused only on screens made of round wires forming a square mesh, as these are defined geometrically alone by the wire diameter d and the porosity

$$\beta = \frac{\text{area of holes}}{\text{total area}} = \left(1 - \frac{d}{l}\right)^2 \quad . \quad . \quad . \quad . \quad (1)$$

where l denotes the width of the square mesh.

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The resistance coefficient k is defined as the pressure drop through the screen divided by the pitot head in the straight tube, *i.e.*

$$k = \frac{p_1 - p_2}{\frac{\rho}{2} u_1^2} \cdot \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad (2)$$

The proposal has been made[1] to reduce the resistance of the screen to that of a single wire, *i.e.* an infinite cylinder. As a fraction of $(1 - \beta)$ of the whole area is covered by pieces of wire with a drag coefficient c, the resistance of the screen is expected to be roughly equal to the sum of the drag of all these wire pieces, or

$$k = (1 - \beta) c. \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad (3)$$

In the same way as c depends on the Reynolds number $u_1 d/\nu$, $k/(1 - \beta)$ should be a function of $u_1 d/\nu$ alone for all screens.

The experimental results, since they give various curves for various values of β, fail to substantiate this. The reason appears to be that the characteristic velocity for the flow round the wires is not the one in front of the screen u_1, but the higher average velocity through the screen, u_1/β. This idea has also been used by G. I. Taylor[5].

Hence, for computing the drag it may be assumed that the single wires are in a field of flow with free stream velocity u_1/β. This may be a bold assumption because the actual velocity in front of the wires is u_1; on the other hand, it is the velocity (outside the boundary layer) on the rear part of the cylinder which influences the wake and therefore the drag. With this assumption the resistance per unit area becomes

$$(1 - \beta) c \frac{\rho}{2} \left(\frac{u_1}{\beta} \right)^2 = k \frac{\rho}{2} u_1^2, \quad . \quad . \quad (4)$$

so that

$$c = \frac{\beta^2}{1 - \beta} k. \quad . \quad . \quad . \quad . \quad . \quad (5)$$

Now c should depend only on $u_1 d/(\beta\nu)$. That this is so is shown in Figs. 1 and 2, where all the available tests are plotted on logarithmic scales. Although k varies between 0·35 and 19, all the measurements lie along a single curve, particularly the very accurate tests from Ref. 6 where, incidentally, an empirical formulae

$$k \simeq \frac{(1 - \beta)}{\beta^2} \quad . \quad . \quad . \quad . \quad . \quad . \quad (6)$$

is given for high Reynolds numbers.

Of this whole series from Ref. 6 only the points marked (O) for a single screen are out of line, but in this case the mesh was rectangular with sides in the ratio 1:1·25,

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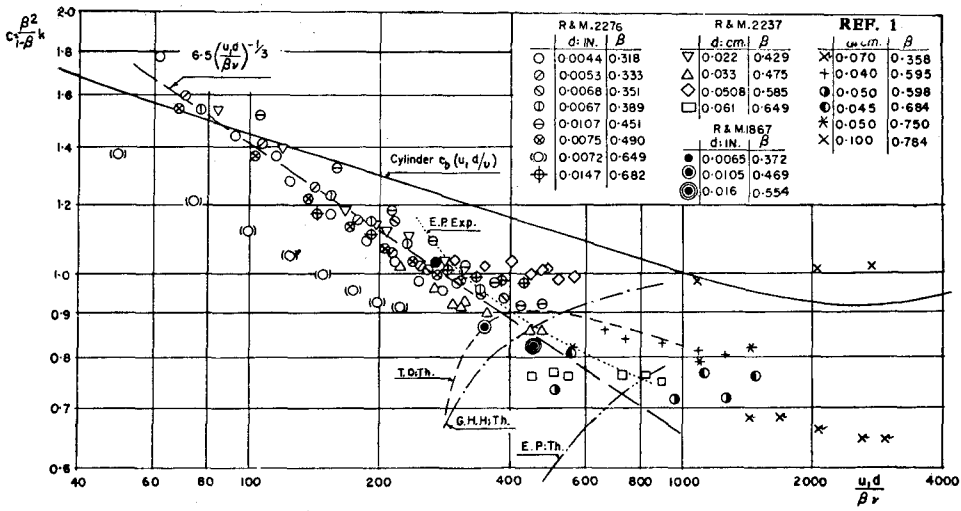


Fig. 1.

The reduced resistance of gauzes, from Refs. 1, 3, 5, 6.

while in all other cases the mesh was square; the discrepancy presumably indicates the influence of the geometry of the system. Other points which do not fit well are the highest values of Ref. 1 (marked $\times$). No explanation can be offered here, and it might be useful to make some new tests at high Reynolds numbers, *e.g.* in a water tunnel. In a water tunnel with a large working section it would be possible to achieve even turbulent flow round the wires (critical Reynolds number $u_1 d/(\beta\nu)$ about 3×10^5).

Further, it is seen from Fig. 1 that the difference between the average c and the drag coefficient c_D of a single cylinder in infinite flow is unexpectedly small, in view of the crudeness of the whole reduction. In particular no attempt has been made to take account of the fact that the wires composing a mesh are not straight, nor of the influence of the points at which wires cross one another.

2. Comparison with Previous Theories

Another theoretical relationship for k has been derived in Ref. 5. Apparently the main interest lay in the resistance of perforated sheets. Hence the flow through such a sheet or a screen was reduced to that through a single hole as at the vena contracta. This gives

$$k=\frac{1-\gamma}{\beta^2 C}-1, \qquad . \quad . \quad . \quad . \quad . \quad (7)$$

with C = contraction coefficient and γ = fraction of the pressure loss which is regained when the stream again becomes uniform behind the sheet. Theoretically this approach is as good as the previous one; but when considerations are restricted to

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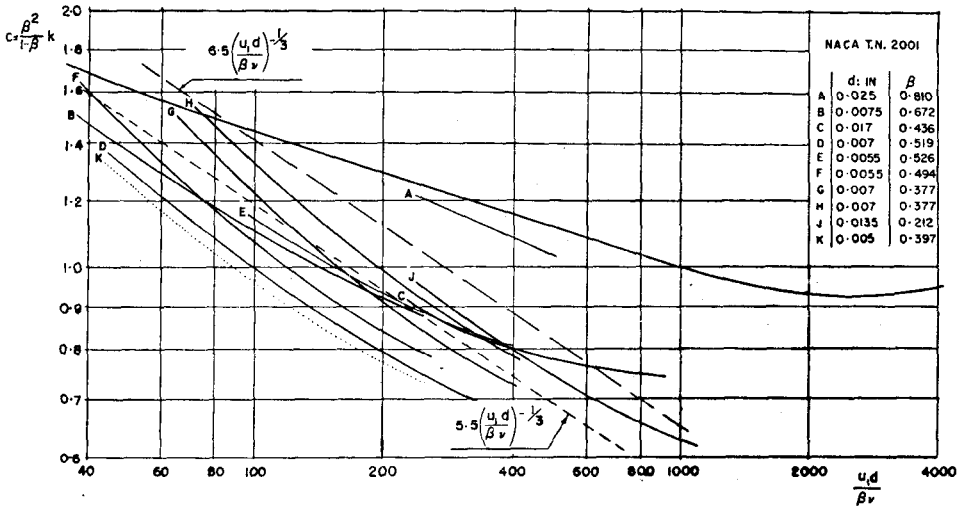


Fig. 2.
The reduced resistance of gauzes, from Ref. 7.

screens made of round wire it seems more practical to take as basis the drag of a cylinder than to find out how C and γ depend on β and the Reynolds number. Furthermore, with most screens, when β is not too small, the mesh length or the distance between the wires is several times the wire diameter. Therefore the comparison with the flow around a cylinder in an infinite fluid seems more natural than that with the jet through a single hole. For screens, it was found that for high Reynolds numbers

$$k=\frac{0{\cdot}70}{\beta^2}-1. \quad . \quad . \quad . \quad . \quad . \quad . \quad (8)$$

Assuming an average value of $u_1 d/\nu$ of 200, this relation gives the curve "T.D : Th." in Fig. 1. (For other values of $u_1 d/\nu$ the same curve has to be moved horizontally.)

From a paper of Eckert and Pflüger[4] a graph of k against β for $u_1 d/\nu = 200$ is also plotted in Fig. 1, giving the curves "E.P : Exp." and "E.P : Th." Their theoretical formulae is

$$k=\left(\frac{1-\beta}{\beta}\right)^2 \quad . \quad . \quad . \quad . \quad . \quad . \quad (9)$$

In Fig. 1 this curve "E.P : Th." does not fit the tests at all whereas in their plotting the agreement is reasonable, though not very good; a striking example of how much depends on the art of presentation.

Glauert, Hirst and Hartshorn[2] derived the formula

$$k=(1-\beta)\left(\frac{1}{\beta^2}-\frac{2}{3}\right) \quad . \quad . \quad . \quad . \quad . \quad (10)$$

which gives the curve "G.H.H:Th." in Fig. 1 again assuming $u_1 d/\nu = 200$.

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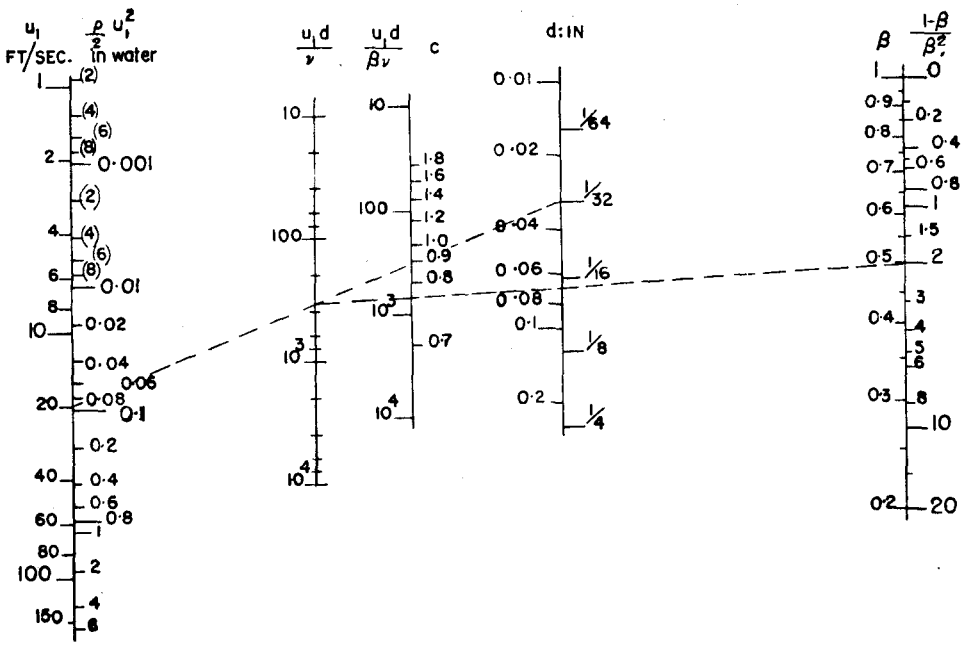


Fig. 3.
Nomogram for the pressure drop in a gauze in air flow.

Figure 1 suggests another formula. For $u_1 d/(\beta\nu)$ between 60 and 600 all the tests at various Reynolds numbers and porosities can be approximated by the broken line

$$c = 6{\cdot}5\left[\left(\frac{u_1 d}{\beta\nu}\right)\right]^{-\frac{1}{3}} \quad . \quad . \quad . \quad . \quad . \quad (11)$$

On the other hand, the American tests[7] in Fig. 2 give lower values,

$$c = 5{\cdot}5\left[\left(\frac{u_1 d}{\beta\nu}\right)\right]^{-\frac{1}{3}} \quad . \quad . \quad . \quad . \quad . \quad (12)$$

No explanation for this difference is obvious. Admittedly the test arrangements were not quite the same. At the National Physical Laboratory the screens were put into a tube, while at the National Bureau of Standards the screens were fixed at the end of a square duct discharging into free air. Still, the flow through the screen should have been the same. As there might have been a difference in the surface roughness of the wires forming the gauzes, it would have been useful to exchange the gauzes of the two test stations. Incidentally, the dotted line K in Fig. 2 corresponds to a silk cloth manufactured for bolting flour.

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Summarising, an average of all the tests may be assumed from Figs. 1 and 2 to be

$$c \simeq 6\left(\frac{u_1 d}{\beta\nu}\right)^{-\frac{1}{3}} \quad . \quad . \quad . \quad . \quad . \quad . \quad (13)$$

or

$$k \simeq 6(1-\beta)\beta^{-\frac{4}{3}}\left(\frac{u_1 d}{\nu}\right)^{-\frac{1}{3}} \quad . \quad . \quad . \quad (14)$$

This formula has been used in the simple nomogram in Fig. 3 for calculating the pressure drop of a gauze in air flow.

For example, if $d=\frac{1}{32}$ in., $\beta=\frac{1}{2}$, $u_1=20$ ft./sec.

$$\frac{u_1 d}{\nu} \simeq 350,$$

$$c \simeq 0{\cdot}78, \frac{\rho}{2}u_1^2 = 0{\cdot}1,$$

and the pressure drop in the gauze is

$$p_1 - p_2 = \frac{\rho}{2}u_1^2 \times \frac{1-\beta}{\beta^2} \times c$$
$$= 0{\cdot}1 \times 2 \times 0{\cdot}78 = 0{\cdot}16 \text{ in. water.}$$

3. Critical Reynolds Number for Eddy Production

Schubauer, Spangenberg and Klebanoff[7] also observed by a hot wire the eddy shedding immediately after the screen and gave a graph for the critical Reynolds number, for the beginning of eddy shedding or of the initial production of turbulence, against β. Obviously a straight line is a good approximation. Yet this gives a constant critical Reynolds number if it is referred again to the velocity u_1/β in the screen:

$$\left(\frac{u_1 d}{\beta\nu}\right)_{\text{crit.}} = 85. \quad . \quad . \quad . \quad . \quad . \quad (15)$$

A Reynolds number of the same order is the smallest at which a vortex street is formed past a single cylinder in infinite fluid as is to be seen from F. Homann's photographs of the wake[8].

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